Analyze Toolwindow Usage Data

2025.09.27 Created by Konstantin Beliakov

Objective

Determine whether there is a statistically significant difference in the time users spend in IDE toolwindows opened automatically versus manually. The provided dataset contains "messy" data, which must be properly processed.

Handling messy data and episode matching strategy

The program iterates through events in the dataset chronologically, tracking for each user whether the window is currently open, as well as how and when it was opened. When a window close event is detected, it can then be easily matched to the corresponding open event.

This approach correctly handles the following cases:

- Unclosed windows (statistics are updated only upon closing)
- Close events without a corresponding open event in the dataset (the close will be processed if the user's state dictionary contains an open record)
- Repeated open events (a new open overwrites the old record)
- Later, we discard sessions longer than 5 hours, partly to exclude invalid data (cases where the user opened a window and did not use it, e.g., walked away from the computer).

Basic statistics

Auto openings: 297

Manual openings: 173

Median time for auto openings: 44,008,344.0 (12h 13m 28s)

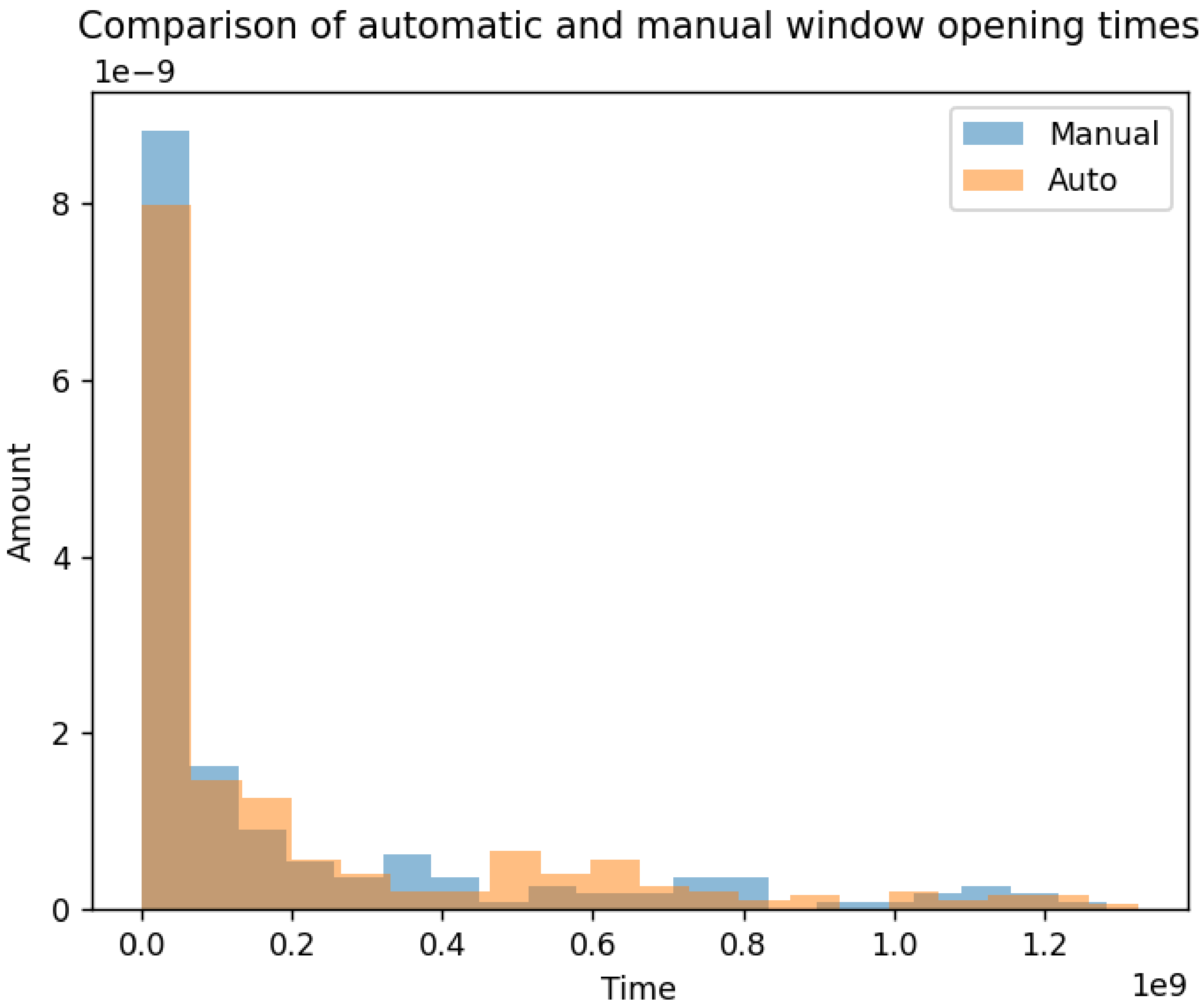
Median time for manual openings: 25,599,895.0 (7h 6m 39s)

Average time for auto openings: 201,494,718.76 (2d 7h 58m 14s)

Average time for manual openings: 188,773,478.72 (2d 4h 26m 13s)

Visualisation

It is difficult to analyze the histogram for the entire sample without statistical tools. But the shortest openings are mostly manual.



Statistical Analysis

The distributions of opening durations in both samples (automatic and manual) do not appear normal. Therefore, it is appropriate to use the **Mann-Whitney U-test** — a standard non-parametric test for comparing two independent samples, checking whether their distributions differ overall.

Mann-Whitney p-value: 0.0767963310424145 (> 0.05)

There are no statistically significant differences.

The mean is not suitable for analyzing our data, since it can be heavily skewed by outliers (e.g., a window left open for 50 hours can shift the average across the sample by 10 minutes). A better approach is to compare medians of the two samples.

Permutation test p-value: 0.7090581883623275

There are no statistically significant differences.

The very high variance has a strongly negative effect on statistical reliability. The p-value is around 0.71, which far exceeds the threshold of 0.05.

Analysis of sessions lasting no more than 5 hours

It can be assumed that for long sessions, it doesn't matter whether the window was opened automatically or manually. Instead, we can analyze only sessions lasting up to 5 hours and perform statistical analysis on this reduced dataset. Shorter sessions are more important: long sessions imply the window is useful and being worked with, while short sessions may be either useful or accidental.

The sample size decreases by 2.2×, but remains sufficiently large.

Analysis for sessions up to 5 hours:

Auto openings: 131

Manual openings: 81

Median time for auto openings: 928,524.0 (15m 28s)

Median time for manual openings: 77,894.0 (1m 17s)

Average time for auto openings: 3,245,547.86 (54m 5s)

Average time for manual openings: 2,001,893.51 (33m 21s)

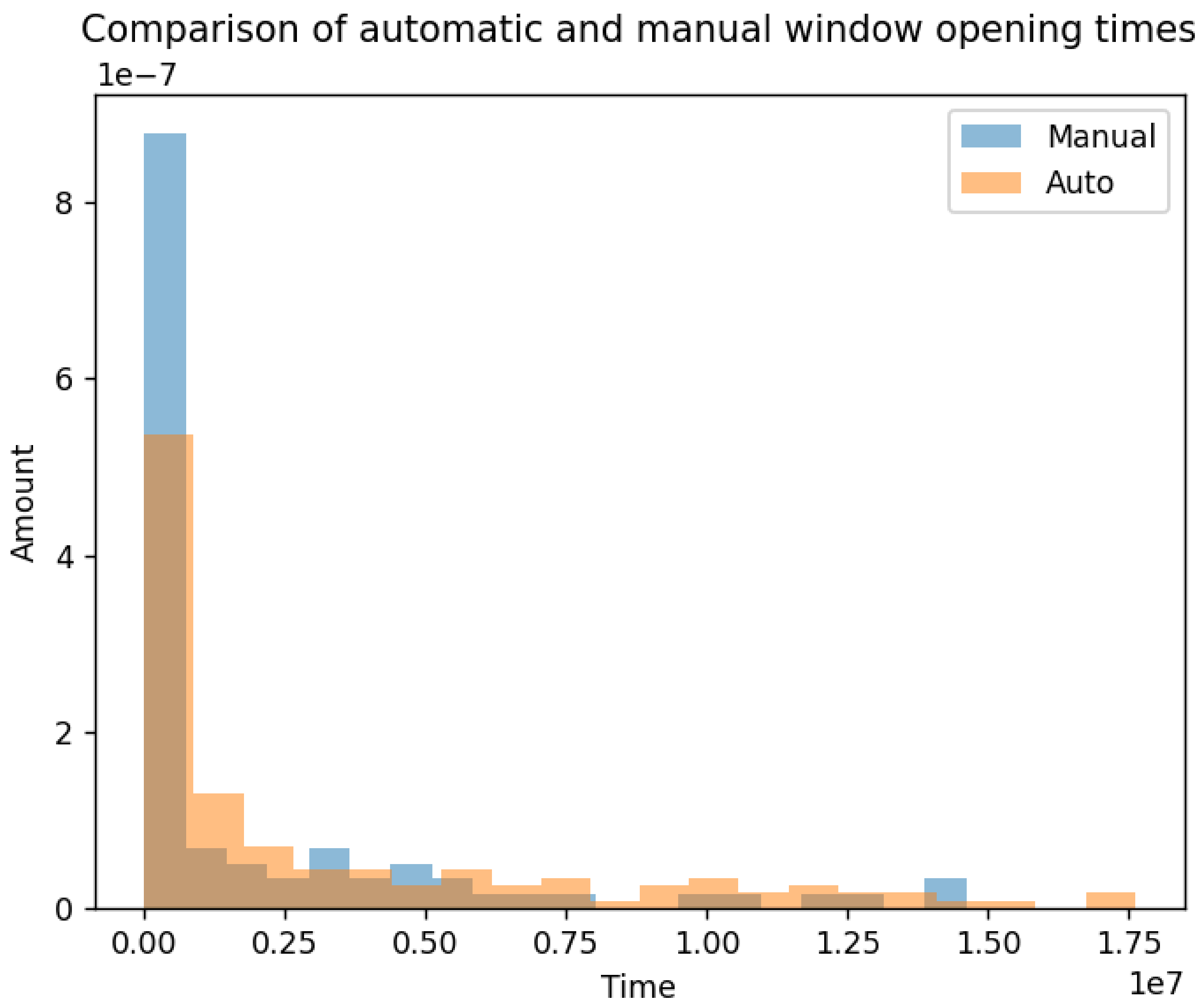
Mann-Whitney p-value: 0.00038072436069249105

The differences are statistically significant ($p < 0.05$)

Median permutation test p-value: 0.03259348130373925

The differences are statistically significant ($p < 0.05$)

Visualisation (for duration less than 5 hours)



From the chart, it can be seen that the shortest window sessions mostly correspond to manual openings, while long usage is characteristic of automatic openings.

Conclusion

In the full dataset, differences are obscured by rare, extremely long sessions that “wash out” the statistics.

However, after removing outliers and keeping only typical usage scenarios (sessions up to 5 hours), it becomes clear that automatically opened toolwindows remain active significantly longer on average.

This makes sense: automatic opening is often tied to important events, such as a test failure or debugger launch, so the user studies it longer than a toolwindow opened manually.